BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 083 · Binary Search

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Does linear search require a sorted list?

Q2. Which is a physical security measure?
A. A strong password
B. File compression
C. A locked server room
D. A software licence

Q3. “The Data Protection Act permits unlimited retention and sharing of all pupil data.” Correct this. Explain why.

Q4. A pupil opens another pupil's account without permission but changes nothing. “Access is legal if no files are deleted.” Correct this. Why could it still break the law?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Binary search — Search method repeatedly checking the midpoint of a sorted range.

Sorted — Arranged in order, such as ascending values.

Low index — Lowest index still being searched.

High index — Highest index still being searched.

Midpoint — Central index of the current search interval.

DIV — Integer quotient operator used to find a whole-number midpoint.

New learning

Sorted prerequisite: Binary search relies on sorted data. Without sorting, discarding half the list may incorrectly discard the target.

Find the midpoint: Keep low and high indices of the remaining range; calculate mid = (low + high) DIV 2 using integer division.

Discard half appropriately: If target is greater than the midpoint value, set low = mid + 1. If smaller, set high = mid - 1.

Repeat until found or empty: Compare the midpoint again until target is found, or low becomes greater than high, indicating no match.

Worked example

A sorted array is [2,4,6,8,10,12,14]. Trace binary search for 12 using low, high and mid. [4 marks]

1. Sorted prerequisite is satisfied; low = 0 and high = 6 [1].

2. mid = (0+6) DIV 2 = 3, value 8 [1].

3. 12 > 8, so discard indexes 0–3; now low = 4, high = 6 [1].

4. mid = (4+6) DIV 2 = 5, value 12, so return index 5 [1].

We Do · Practise with guidance

Guided 1

Trace binary search for 9 in sorted [1,3,5,7,9], noting each midpoint. [4 marks]

Worked method: low 0, high 4 gives mid 2, value 5 [1]. Since 9 > 5, low = 3 [1]. mid (3+4) DIV 2 = 3 gives 7 [1]; low = 4 and index 4 contains 9 [1].

Guided 2

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Q2. Mid = 2, value 8 [1]; target 14 > 8 so low = 3 [1]; next mid = 3, value 10 [1]; target 14 >10 so index 4 is checked and found [1].

Q3. Begin with low 0 high 6 [1], midpoint index 3 value 7 [1]. Target 5 is smaller so reduce high to 2 [1]. New midpoint index 1 value 3, so move low to 2 [1]; index 2 contains 5 [1]. Sorted data is essential because only then can half be discarded safely [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. No; it examines items in sequence until a match is found or the list ends.

Q2. C — A locked server room

Q3. It sets rules for using personal data lawfully and protects people's rights.

Q4. Access without permission can break the Computer Misuse Act, even if no files change. A law or licence sets what is allowed; having a copy does not give unlimited rights.

I Do / We Do pages · Guided practice

Q1. low 0, high 4 gives mid 2, value 5 [1]. Since 9 > 5, low = 3 [1]. mid (3+4) DIV 2 = 3 gives 7 [1]; low = 4 and index 4 contains 9 [1].

Q2. The list is not sorted [1], so discarding half based on a midpoint comparison may wrongly exclude the target [1].

Q3. First mid index 2 gives value 4 [1], so low becomes 3 [1]. Next mid index 3 gives 7 [1], so high becomes 2 and low > high; target is absent [1].

You Do pages · Code First

```
array weights = [1,2,4,8,16]
target = 16
low = 0
high = 4
mid = (low + high) DIV 2
if weights[mid] < target then
    low = mid + 1
endif
print(low)
```

Expected result: 3

You Do pages · Independent questions

Q1. Values must be in sorted order [1].

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Explain what happens if binary search is used on [10,3,8,1] without sorting. [2 marks]

Worked method: The list is not sorted [1], so discarding half based on a midpoint comparison may wrongly exclude the target [1].

Guided 3

Trace search for target 6 in sorted [1,2,4,7,9] and explain the unsuccessful result. [4 marks]

Worked method: First mid index 2 gives value 4 [1], so low becomes 3 [1]. Next mid index 3 gives 7 [1], so high becomes 2 and low > high; target is absent [1].

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | You Do Work independently | Review & Improve Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the assignment so the search moves right after comparing the middle value.

```
array weights = [1,2,4,8,16]
target = 16
low = 0
high = 4
mid = (low + high) DIV 2
if weights[mid] < target then
    ...
endif
print(low)
```

write

Question 1

State the required property of an array before binary search can be used correctly. [1 mark] Source: Original OCR J277-style question; not an official OCR past-paper original.

Question 2

For sorted [2,4,8,10,14], trace binary search for 14, showing each midpoint and discarded half. [4 marks] Source: Original OCR J277-style question; not an official OCR past-paper original.

Question 3

Explain how binary search finds 5 in sorted [1,3,5,7,9,11,13], why sorting is essential and what happens when a target is missing. [6 marks] Source: Original OCR J277-style question; not an official OCR past-paper original.

Exit Ticket · Answer before checking

Binary search works reliably on any unsorted list.

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check understanding

DRAFT · Teacher review required before pupil use Page 5